Time Series

It is an **arrangement** of the statistical data in accordance with the **time of occurrence**.

Most of the series are related to **Economics, Business and Commerce**, e.g., the series related to **prices**, production and consumption of various commodities; **investment, sales and profit of business house**.

Time Series Analysis

A time series is a series of figures recorded over **time**, e.g. unemployment over the last 5 **years**, output over the last 12 **months**, etc.

Time series analysis is a technique used to:

- identify whether there is any **underlying historical trend**
- use this analysis of the historical trend **to forecast the trend into the future**
- identify whether there are any **seasonal variations** around the trend
- apply estimated seasonal variations **to a trend line forecast** in order to prepare a forecast season by season.

Time Series Forecasting

- Time series forecasting occurs **when you make scientific predictions** based on **historical time stamped data**.

- It involves **building models through historical analysis** and using them to **make observations and drive future strategic decision-making**.

- An important distinction in forecasting is that at the **time of the work**, the **future outcome is completely unavailable** and can only be **estimated** through **careful analysis and evidence-based priors**.

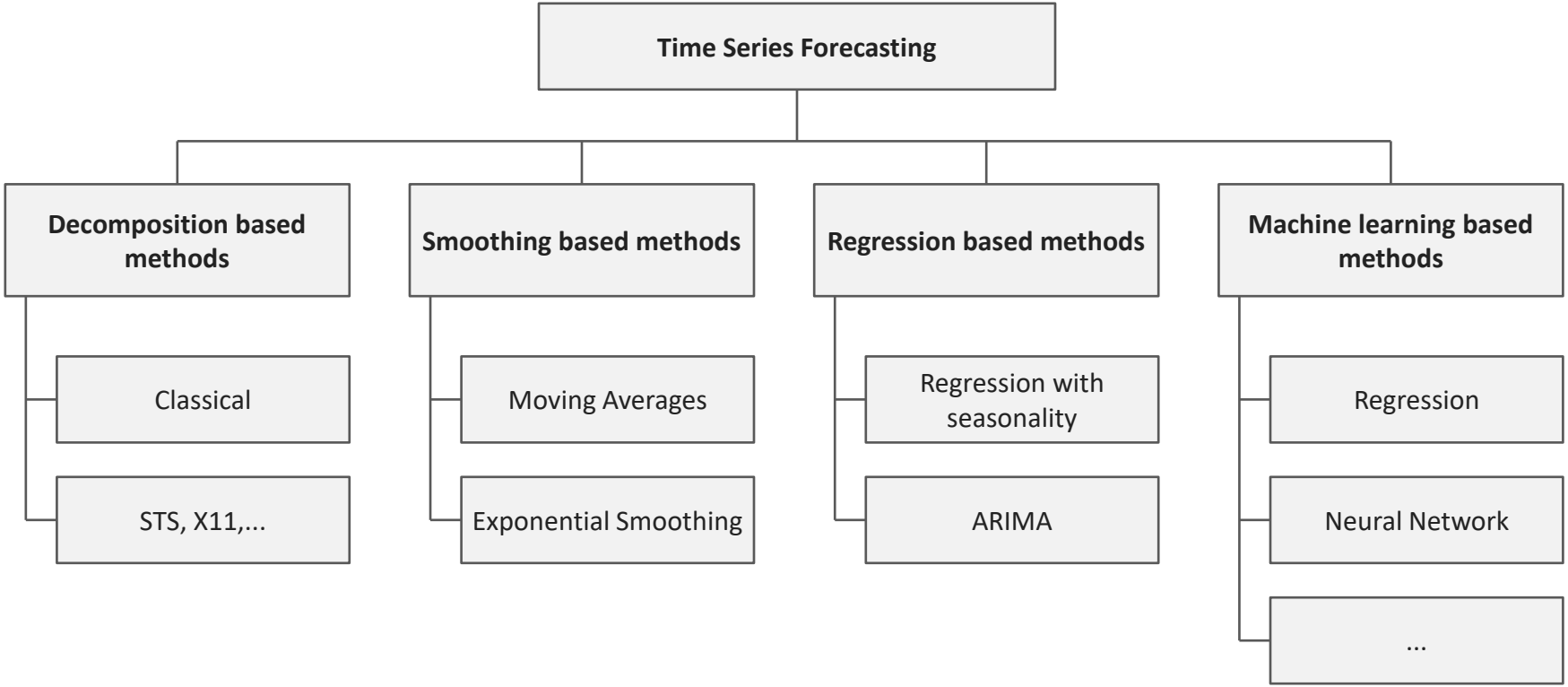
Applications of Time Series Forecasting

Forecasting has a range of applications in various industries:

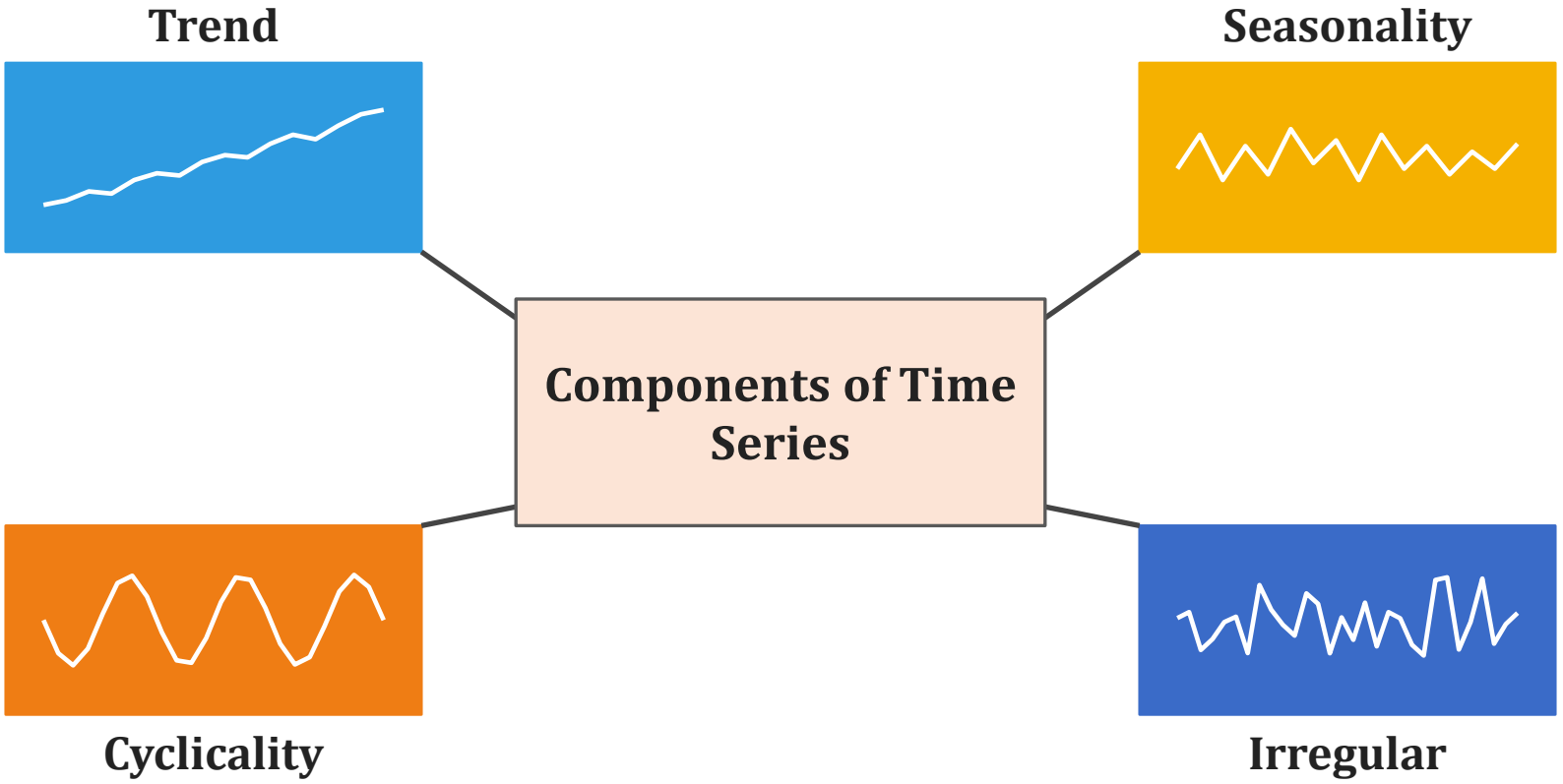
- Weather forecasting
- Economic forecasting
- Engineering forecasting
- Retail forecasting
- Environmental studies forecasting
- Climate forecasting
- Healthcare forecasting
- Finance forecasting
- Business forecasting
- Social studies forecasting

…and more.

Taxonomy of Forecasting Techniques



Components of Time Series



Components of Time Series: Details

Four components make up a time series

Component	Description
Trend	Long-term movement of data showing increase, decrease or stability; can be linear or nonlinear.
Seasonality	Regular patterns repeating at fixed intervals (daily, monthly, yearly), often due to seasons or events.
Cyclical Variations	Long-term fluctuations without fixed periods, linked to economic or business cycles.
Irregularity (Noise)	Random, unpredictable variations caused by events, errors or other unforeseen factors.

Importance of Time Series Analysis

- **Predicting Future Trends:** Helps forecast outcomes like demand, revenue or stock prices.
- **Detecting Patterns and Anomalies:** Identifies cycles, seasonal effects and unusual behaviours.
- **Risk Mitigation:** Provides early warnings to prevent potential losses.
- **Strategic Planning:** Supports budgeting, staffing, inventory and policy decisions.
- **Competitive Advantage:** Enables faster response and optimised operations.

When Should You Use Time Series Forecasting

Use Time Series Forecasting when:

- Historical data is consistent and reliable.
- Temporal patterns or trends exist in the data.
- Future behaviour depends on past observations.

Avoid Time Series Forecasting when:

- Data is highly noisy or erratic.
- No meaningful historical patterns are present.
- Key influencing factors are unknown or unpredictable.

What is Decomposition?

Splitting a time series into its components

- **Definition:** Decomposition breaks a time series into separate components: Trend (T), Seasonality (S), Cyclical (C) and Irregular / Residual (I).
- **How:** Each component is estimated separately, then combined to describe or forecast the series.
- **Why use it:** understand the underlying pattern, remove seasonality (seasonal adjustment), improve forecasts and spot anomalies in the residual.
- **Methods:** Classical, STS and X11 are decomposition-based forecasting methods.

Additive vs Multiplicative Decomposition

Model	Formula	Use when	Example
Additive	$Y(t) = T(t) + S(t) + R(t)$	Seasonal swings stay about the same size	$100 + 15 - 3 = 112$
Multiplicative	$Y(t) = T(t) \times S(t) \times R(t)$	Seasonal swings grow with the level of the series	$100 \times 1.15 \times 0.97 = 111.55$

Classical decomposition steps

- Estimate the trend T with a centered moving average.
- Detrend ($Y - T$ or Y / T), then average by season to get the seasonal component S.
- Residual R = what is left: $Y - T - S$ (additive) or $Y / (T \times S)$ (multiplicative).

Additive Decomposition

$$Y(t) = T(t) + S(t) + R(t)$$

Observed = Trend + Seasonal + Residual

T Trend

Long-term direction, usually estimated with a moving average.

S Seasonal

Repeating pattern of constant size, such as every quarter.

R Residual

What is left over: R = Y − T − S.

Use the additive model when the seasonal swing stays the same size, whether the level is rising or falling.

Worked Example: Quarterly Sales

Three years of quarterly sales (m = 4 seasons). Forecast for Q2 in year 4. And Q3 in year 5

Year	Q1	Q2	Q3	Q4
Year 1	18	20	34	44
Year 2	22	20	38	48
Year 3	18	20	42	56

t	Y(t)
Y1Q1	18
Y1Q2	20
Y1Q3	34
Y1Q4	44
Y2Q1	22
Y2Q2	20
Y2Q3	38
Y2Q4	48
Y3Q1	18
Y3Q2	20
Y3Q3	42
Y3Q4	56

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

t	Y(t)	T(t)
Y1Q1	18	—
Y1Q2	20	—
Y1Q3	34	**29.5**
Y1Q4	44	
Y2Q1	22	
Y2Q2	20	
Y2Q3	38	
Y2Q4	48	
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 18 + 20 + 34 + 44 + \tfrac{1}{2}\cdot 22) \div 4 = 29.5$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

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Y2Q1	22	
Y2Q2	20	
Y2Q3	38	
Y2Q4	48	
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 20 + 34 + 44 + 22 + \tfrac{1}{2}\cdot 20) \div 4 = 30$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

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t	Y(t)	T(t)
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Y1Q3	34	**29.5**
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Y2Q4	48	
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 34 + 44 + 22 + 20 + \tfrac{1}{2}\cdot 38) \div 4 = 30.5$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

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Y1Q1	18	—
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Y1Q3	34	**29.5**
Y1Q4	44	**30**
Y2Q1	22	**30.5**
Y2Q2	20	**31.5**
Y2Q3	38	
Y2Q4	48	
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 44 + 22 + 20 + 38 + \tfrac{1}{2}\cdot 48) \div 4 = 31.5$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

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Y1Q3	34	**29.5**
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Y2Q1	22	**30.5**
Y2Q2	20	**31.5**
Y2Q3	38	**31.5**
Y2Q4	48	
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 22 + 20 + 38 + 48 + \tfrac{1}{2}\cdot 18) \div 4 = 31.5$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

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Y2Q1	22	**30.5**
Y2Q2	20	**31.5**
Y2Q3	38	**31.5**
Y2Q4	48	**31.0**
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: T(3) = (½·20 + 38 + 48 + 18 + ½·20) ÷ 4 = 31.0

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

t	Y(t)	T(t)
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Y1Q2	20	—
Y1Q3	34	**29.5**
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Y2Q2	20	**31.5**
Y2Q3	38	**31.5**
Y2Q4	48	**31.0**
Y3Q1	18	**31.5**
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 38 + 48 + 18 + 20 + \tfrac{1}{2}\cdot 42) \div 4 = 31.5$

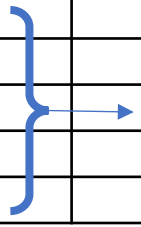
An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

t	Y(t)	T(t)
Y1Q1	18	—
Y1Q2	20	—
Y1Q3	34	**29.5**
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Y2Q2	20	**31.5**
Y2Q3	38	**31.5**
Y2Q4	48	**31.0**
Y3Q1	18	**31.5**
Y3Q2	20	**33**
Y3Q3	42	—
Y3Q4	56	—



Example, t = 3: T(3) = (½·48 + 18 + 20 + 42 + ½·56) ÷ 4 = 33

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Additive Step 1: Detrending

Estimate Detrending

Detrending= Y(t) -T(t)

t	Y(t)	T(t)	Y(t)-T(t)
Y1Q1	18	—	—
Y1Q2	20	—	—
Y1Q3	34	**29.5**	**4.5**
Y1Q4	44	**30**	**14**
Y2Q1	22	**30.5**	**-8.5**
Y2Q2	20	**31.5**	**-11.5**
Y2Q3	38	**31.5**	**6.5**
Y2Q4	48	**31.0**	**17**
Y3Q1	18	**31.5**	**-13.5**
Y3Q2	20	**33**	**-13**
Y3Q3	42	—	—
Y3Q4	56	—	—

Additive Step 2: Seasonal Indices

t	Y(t)	T(t)	Y(t)-T(t)
Y1Q1	18	—	—
Y1Q2	20	—	—
Y1Q3	34	29.5	4.5
Y1Q4	44	30	14
Y2Q1	22	30.5	-8.5
Y2Q2	20	31.5	-11.5
Y2Q3	38	31.5	6.5
Y2Q4	48	31.0	17
Y3Q1	18	31.5	-13.5
Y3Q2	20	33	-13
Y3Q3	42	—	—
Y3Q4	56	—	—

Quarter	Q1	Q2	Q3	Q4
Year1	—	—	4.5	14
Year2	-8.5	-11.5	6.5	17
Year3	-13.5	-13.0	—	—
Total	-22	-24.5	11	31
Average	-11	-12.25	5.5	15.5

Additive Step 2: Seasonal Indices

Seasonal indices must sum to 0

Adjustment = sum ÷ 4 = −2.25 ÷ 4 = −0.5625. Subtract it, i.e. add 0.5625 to each.

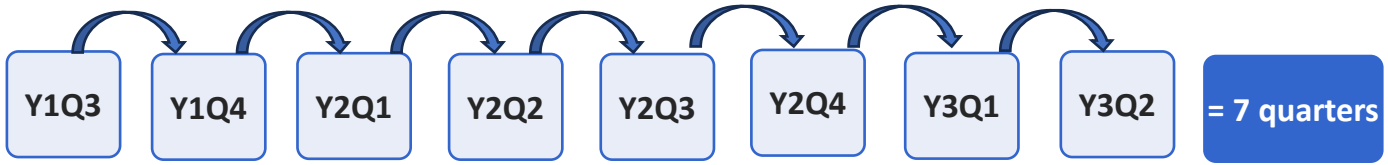
	Q1	Q2	Q3	Q4	Sum
Raw average	−11	−12.25	5.5	15.5	**−2.25**
Adjustment	+0.5625	+0.5625	+0.5625	+0.5625	+2.25
Seasonal index S	**−10.4375**	**−11.6875**	**6.0625**	**16.0625**	**0**

Meaning: Q1 and Q2 run below trend, Q3 and Q4 above it (Q4 by about 16).

Additive Step 3: Trend per Quarter

How fast does the trend grow?

Trend per quarter = (last T – first T) ÷ number of quarters between them



Point	Quarter	t	T(t)
First	Y1Q3	3	**29.5**
Last	Y3Q2	10	**33**

(33 – 29.5) ÷ (10-3)

= 0.5

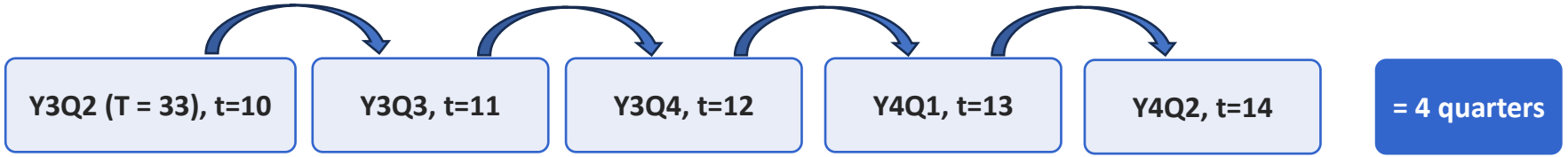
t		Y(t)	T(t)	Y(t)-T(t)
1	Y1Q1	18	—	—
2	Y1Q2	20	—	—
3	Y1Q3	34	**29.5**	**4.5**
4	Y1Q4	44	**30**	**14**
5	Y2Q1	22	**30.5**	**-8.5**
6	Y2Q2	20	**31.5**	**-11.5**
7	Y2Q3	38	**31.5**	**6.5**
8	Y2Q4	48	**31.0**	**17**
9	Y3Q1	18	**31.5**	**-13.5**
10	Y3Q2	20	**33**	**-13**
11	Y3Q3	42	—	—
12	Y3Q4	56	—	—

Between t = 3 and t = 10 there are 7 quarters. The deseasonalized sales rise by 0.5 each quarter.

Additive Step 4: Forecast Q2 of Year 4

Forecast = Trend + Seasonal index

Count the quarters from the last trend value (Y3Q2):

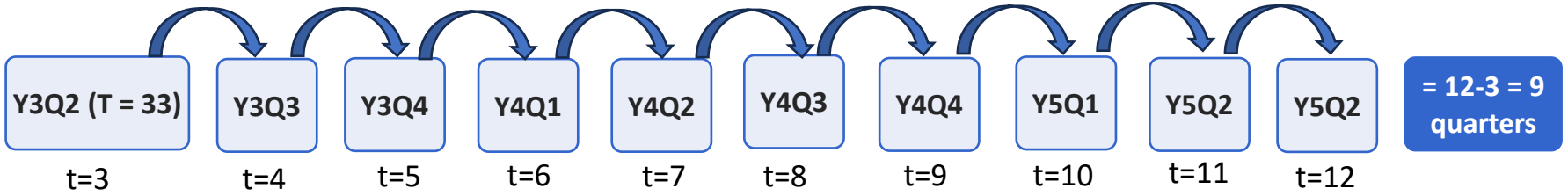


Step	Calculation	Value
Trend at Y3Q2	from central moving average	33
Increase in quarters	14-10	4
Increase over 4 quarters	4 × 0.5	+2
Trend at Y4Q2	33 + 2	**35**
Seasonal index Q2	adjusted S	−11.6875
Forecast	35 − 11.6875	**23.31**

Additive Step 4: Forecast Q3 of Year 5

Forecast = Trend + Seasonal index

Count the quarters from the last trend value (Y3Q2):



Step	Calculation	Value
Trend at Y3Q2	from moving average	33
Increase over 9 quarters	9 × 0.5	+4.5
Trend at Y5Q3	33 + 4.5	**37.5**
Seasonal index Q3	adjusted S	+6.0625
Forecast	37.5 + 6.0625	**43.56**

Multiplicative Decomposition

$Y(t) = T(t) \times S(t) \times R(t)$

Observed = Trend x Seasonal x Residual

T Trend

Long-term direction, usually estimated with a moving average.

S Seasonal

Repeating pattern of constant size, such as every quarter.

R Residual

What is left over: R = Y − T − S.

Use the additive model when the seasonal swing stays the same size, whether the level is rising or falling.

Worked Example: Quarterly Sales

Three years of quarterly sales (m = 4 seasons). Forecast for Q2 in year 4. And Q3 in year 5

Year	Q1	Q2	Q3	Q4
Year 1	18	20	34	44
Year 2	22	20	38	48
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t	Y(t)
Y1Q1	18
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Multiplicative Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

t	Y(t)	T(t)
Y1Q1	18	—
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Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 18 + 20 + 34 + 44 + \tfrac{1}{2}\cdot 22) \div 4 = 29.5$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

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Y2Q2	20	
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Y2Q4	48	
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Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 20 + 34 + 44 + 22 + \tfrac{1}{2}\cdot 20) \div 4 = 30$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

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Estimate T with a centered 2×4 moving average

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Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

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Y2Q3	38	
Y2Q4	48	
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: T(3) = (½·44 + 22 + 20 + 38 + ½·48) ÷ 4 = 31.5

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Multiplicative Step 1: Trend

Estimate T with a centered 2×4 moving average

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Y2Q3	38	**31.5**
Y2Q4	48	
Y3Q1	18	
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 22 + 20 + 38 + 48 + \tfrac{1}{2}\cdot 18) \div 4 = 31.5$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Multiplicative Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

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Y3Q1	18	
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Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 20 + 38 + 48 + 18 + \tfrac{1}{2}\cdot 20) \div 4 = 31.0$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Multiplicative Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

t	Y(t)	T(t)
Y1Q1	18	—
Y1Q2	20	—
Y1Q3	34	**29.5**
Y1Q4	44	**30**
Y2Q1	22	**30.5**
Y2Q2	20	**31.5**
Y2Q3	38	**31.5**
Y2Q4	48	**31.0**
Y3Q1	18	**31.5**
Y3Q2	20	
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: $T(3) = (\tfrac{1}{2}\cdot 38 + 48 + 18 + 20 + \tfrac{1}{2}\cdot 42) \div 4 = 31.5$

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Multiplicative Step 1: Trend

Estimate T with a centered 2×4 moving average

$$T(t) = [\ \tfrac{1}{2}\cdot Y(t-2) + Y(t-1) + Y(t) + Y(t+1) + \tfrac{1}{2}\cdot Y(t+2)\ ] \div 4$$

t	Y(t)	T(t)
Y1Q1	18	—
Y1Q2	20	—
Y1Q3	34	**29.5**
Y1Q4	44	**30**
Y2Q1	22	**30.5**
Y2Q2	20	**31.5**
Y2Q3	38	**31.5**
Y2Q4	48	**31.0**
Y3Q1	18	**31.5**
Y3Q2	20	**33**
Y3Q3	42	—
Y3Q4	56	—

Example, t = 3: T(3) = (½·48 + 18 + 20 + 42 + ½·56) ÷ 4 = 33

An even season length (4) needs the 2×4 average so the result is centered on t. The first and last two quarters get no trend value.

Multiplicative Step 1: Detrending

Estimate Detrending

Detrending= Y(t) / T(t)

t	Y(t)	T(t)	Y(t)/T(t)
Y1Q1	18	—	—
Y1Q2	20	—	—
Y1Q3	34	29.5	1.153
Y1Q4	44	30	1.467
Y2Q1	22	30.5	0.721
Y2Q2	20	31.5	0.635
Y2Q3	38	31.5	1.206
Y2Q4	48	31.0	1.548
Y3Q1	18	31.5	0.571
Y3Q2	20	33	0.606
Y3Q3	42	—	—
Y3Q4	56	—	—

Multiplicative Step 2: Seasonal Indices

t	Y(t)	T(t)	Y(t)/T(t)
Y1Q1	18	—	—
Y1Q2	20	—	—
Y1Q3	34	29.5	1.153
Y1Q4	44	30	1.467
Y2Q1	22	30.5	0.721
Y2Q2	20	31.5	0.635
Y2Q3	38	31.5	1.206
Y2Q4	48	31.0	1.548
Y3Q1	18	31.5	0.571
Y3Q2	20	33	0.606
Y3Q3	42	—	—
Y3Q4	56	—	—

Quarter	Q1	Q2	Q3	Q4
Year1	—	—	1.153	1.467
Year2	0.721	0.635	1.206	1.548
Year3	0.571	0.606	—	—
Total	1.292	1.241	1.206	1.548
Average	0.646	0.6205	1.1795	1.5075

Multiplicative Step 2: Seasonal Indices

Seasonal indices must sum to 4

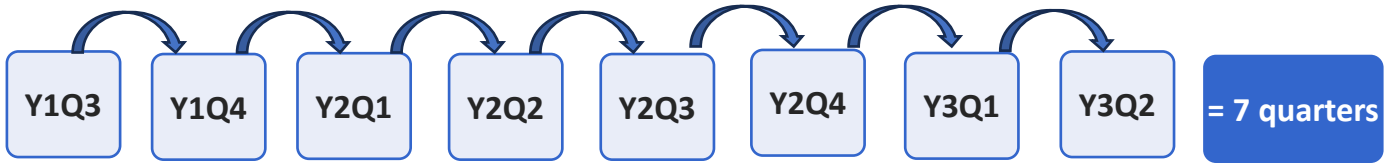
Adjustment = sum ÷ 4 = 3.9535 ÷ 4 = 0.9884. Divide each by 0.9884.

	Q1	Q2	Q3	Q4	Sum
Raw average	0.646	0.6205	1.1795	1.5075	**3.9535**
Adjustment	0.9884	0.9884	0.9884	0.9884	**3.9535**
Seasonal index S	**0.654**	**0.628**	**1.193**	**1.525**	**4**

Multiplicative Step 3: Trend per Quarter

How fast does the trend grow?

Trend per quarter = (last T – first T) ÷ number of quarters between them



Point	Quarter	t	T(t)
First	Y1Q3	3	**29.5**
Last	Y3Q2	10	**33**

(33 – 29.5) ÷ (10-3)

= 0.5

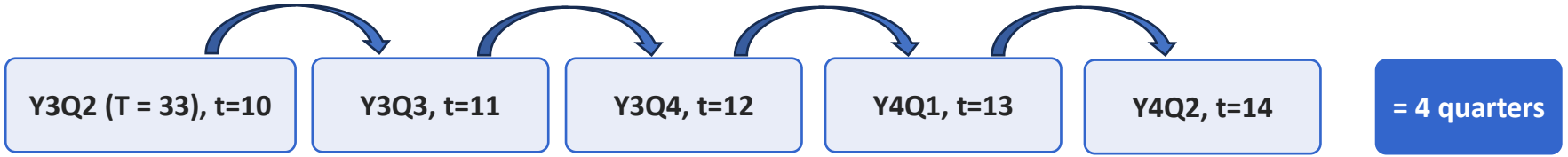
t		Y(t)	T(t)	Y(t)/T(t)
1	Y1Q1	18	—	—
2	Y1Q2	20	—	—
3	Y1Q3	34	**29.5**	**1.153**
4	Y1Q4	44	**30**	**1.467**
5	Y2Q1	22	**30.5**	**0.721**
6	Y2Q2	20	**31.5**	**0.635**
7	Y2Q3	38	**31.5**	**1.206**
8	Y2Q4	48	**31.0**	**1.548**
9	Y3Q1	18	**31.5**	**0.571**
10	Y3Q2	20	**33**	**0.606**
11	Y3Q3	42	—	—
12	Y3Q4	56	—	—

Between t = 3 and t = 10 there are 7 quarters. The deseasonalized sales rise by 0.5 each quarter.

Multiplicative Step 4: Forecast Q2 of Year 4

Forecast = Trend x Seasonal index

Count the quarters from the last trend value (Y3Q2):

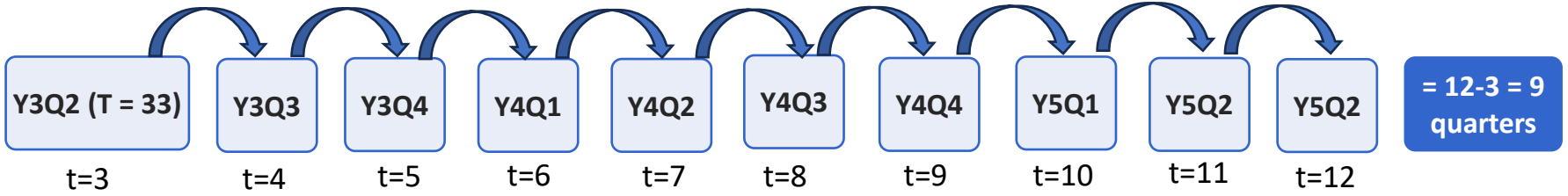


Step	Calculation	Value
Trend at Y3Q2	from central moving average	33
Increase quarters	14-10	4
Increase over 4 quarters	4 × 0.5	+2
Trend at Y4Q2	33 + 2	**35**
Seasonal index Q2	adjusted S	0.628
Forecast	35 x 0.628	**21.98**

Multiplicative Step 4: Forecast Q3 of Year 5

Forecast = Trend x Seasonal index

Count the quarters from the last trend value (Y3Q2):



Step	Calculation	Value
Trend at Y3Q2	from moving average	33
Increase over 9 quarters	9 × 0.5	+4.5
Trend at Y5Q3	33 + 4.5	**37.5**
Seasonal index Q3	adjusted S	1.193
Forecast	37.5 x 1.193	**44.75**

Smoothing Based Methods

In smoothing based approaches, an observation is a **function of past few observations**.

Used to forecast the next value of a time series, e.g., **sales, prices, demand**.

Notation

Symbol	Meaning
$t = 1, 2, \ldots, n$	Time periods (days, months, years…)
$y_1, y_2, \ldots, y_n$	Observed data for each period
F_{n+h}	Forecast for h periods after n (usually h = 1)
$e_t = y_t - F_t$	Forecast error at time t

Performance Evaluation

Mean Absolute Error (MAE): MAE measures the average magnitude of errors in a set of predictions, without considering their direction (treats all errors equally).

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|e_t|$$

Root Mean Square Error (RMSE): RMSE is the square root of the average of squared differences between predicted and actual values.

$$(\mathrm{RMSE} = \sqrt{\mathrm{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}|e_t|^2})$$

Performance Evaluation

Mean Absolute Percentage Error (MAPE): MAPE measures errors as a percentage of the actual values, making it scale-independent and easy to compare across different datasets.

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{e_t}{y_i}\right|$$

Mean Absolute Scaled Error (MASE): MASE (proposed by Hyndman and Koehler) scales the model's MAE against the in-sample MAE of a simple naïve (or baseline) forecast. A MASE (<1) means the model performs better than the naïve baseline, while MASE (>1) means it performs worse.

$$\text{MASE} = \frac{\text{MAE}}{\text{MAE}_{\text{in-sample, naïve}}} = \frac{\frac{1}{n}\sum_{i=1}^{n}|e_t|}{\frac{1}{n-1}\sum_{j=2}^{n}|y_j - y_{j-1}|}$$

Average Method

Forecast = average of all past observations

$$F(t) = (y(1) + y(2) + \ldots + y(t-1)) \div (t-1) \text{ and } e(t) = y(t) - F(t)$$

t	y(t)	Forecast F(t)	Error e(t) = y(t) – F(t)
1	20	—	—
2	22	F(2) = 20 ÷ 1 = 20.00	e(2) = 22 − 20.00 = 2.00
3	21	F(3) = (20+22) ÷ 2 = 21.00	e(3) = 21 − 21.00 = 0.00
4	25	F(4) = (20+22+21) ÷ 3 = 21.00	e(4) = 25 − 21.00 = 4.00
5	24	F(5) = (20+22+21+25) ÷ 4 = 22.00	e(5) = 24 − 22.00 = 2.00
6	26	F(6) = (20+22+21+25+24) ÷ 5 = 22.40	e(6) = 26 − 22.40 = 3.60
7	**?**	**F(7) = (20+22+21+25+24+26) ÷ 6 = 23.00**	**—**

Error: e(t) = actual − forecast. Positive means the forecast was too low.

Problem: every new point is added, so old data never leaves the average. This makes it slow to react.

Performance Evaluation: Average Method

t	y(t)	F(t)	e(t)	|e(t)|	$e(t)^2$	|e(t)| ÷ y(t)	Naive error |y(t) – y(t–1)|
2	22	20.00	2.00	2.00	4.00	9.09%	22-20 = 2
3	21	21.00	0.00	0.00	0.00	0.00%	1
4	25	21.00	4.00	4.00	16.00	16.00%	4
5	24	22.00	2.00	2.00	4.00	8.33%	1
6	26	22.40	3.60	3.60	12.96	13.85%	2
Sum			**11.60**	**11.60**	**36.96**	**47.27%**	**10**

MAE Average of |e| = 11.60 ÷ 5 **2.32**

RMSE √ (Average of e^2) = √(36.96 ÷ 5) **2.72**

MAPE Average of |e| ÷ y = 47.27% ÷ 5 **9.45%**

MASE MAE ÷ Avg(naive error) = 2.32 ÷ 2.00 **1.16**

Naive error = change from the previous value. Average = 10 ÷ 5 = 2.00

Moving Average Method

Forecast = average of the last k observations (here k = 3)

$$F(t) = (y(t-1) + y(t-2) + y(t-3)) \div 3 \quad \text{and} \quad e(t) = y(t) - F(t)$$

t	y(t)	Forecast F(t)	Error e(t) = y(t) − F(t)
1	20	—	—
2	22	—	—
3	21	—	—
4	25	F(4) = (20+22+21) ÷ 3 = 21.00	e(4) = 25 − 21.00 = 4.00
5	24	F(5) = (22+21+25) ÷ 3 = 22.67	e(5) = 24 − 22.67 = 1.33
6	26	F(6) = (21+25+24) ÷ 3 = 23.33	e(6) = 26 − 23.33 = 2.67
7	**?**	**F(7) = (25+24+26) ÷ 3 = 25.00**	**—**

Problem: the average method keeps every old value, so it is slow to react when the data changes.

Fix: use only the last k = 3 values. Older values drop out of the window.

Limitation: no forecast for the first k points, and k must be chosen.

Performance Evaluation: Moving Average

t	y(t)	F(t)	e(t)	|e(t)|	e(t)²	|e(t)| ÷ y(t)	Naive error |y(t) − y(t−1)|
4	25	21.00	4.00	4.00	16.00	16.00%	4
5	24	22.67	1.33	1.33	1.78	5.56%	1
6	26	23.33	2.67	2.67	7.11	10.26%	2
Sum			**8.00**	**8.00**	**24.89**	**31.81%**	**7**

MAE Average of |e| = 8 ÷ 3 **2.67**

RMSE √ (Average of e²) = √(24.89 ÷ 3) **2.88**

MAPE Average of |e| ÷ y = 31.81% ÷ 3 **10.60%**

MASE MAE ÷ Avg(naive error) = 2.67 ÷ 2.33 **1.14**

Naive error = change from the previous value. Average = 7 ÷ 3 = 2.33

Linear Regression

Least Square Method, Worked Example & Performance Evaluation

Steps in Linear Regression

1. **Data Collection**
   Gather the data for the independent and dependent variable.

2. **Plot the Data**
   Create a scatter plot to visualize the relationship between the variables.

3. **Calculate the Slope & Intercept**
   Compute slope a and intercept b using the least square formulas.

4. **Fit the Regression**
   Use the calculated slope and intercept to fit the best line through the data points.

5. **Make Prediction**
   Use the regression equation to predict the dependent variable for any given independent value.

Calculating Slope and Intercept

Slope (a)

$$a = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

Intercept (b)

$$b = \frac{\sum y - a(\sum x)}{n}$$

Regression line: $y' = ax + b$

Example 1: Expenditure Data

The expenditure of an organization (in thousand) for every month is shown in the table below.

X (Month)	1	2	3	4	5
Y (Expenditure)	12	19	29	37	45

Task

- Find the regression line y = ax + b using the least square method.
- Estimate the expenditure of the company in the 6th month using the line as a model.

Solution (i): Calculate xy and x^2

Sr. No	x	y	xy	x^2
1	1	12	12	1
2	2	19	38	4
3	3	29	87	9
4	4	37	148	16
5	5	45	225	25
Total	**15**	**142**	**510**	**55**

15 Σx	**142** Σy
510 Σxy	**55** Σx^2

n = 5 data points

Solution (ii): Find a and b

Slope

$$a = \frac{5 \times 510 - 15 \times 142}{5 \times 55 - 225} = \frac{420}{50} = 8.4$$

a = 8.4

slope

Intercept

$$b = \frac{142 - 8.4 \times 15}{5} = \frac{16}{5} = 3.2$$

b = 3.2

intercept

Solution (iii): Fit the Regression Line

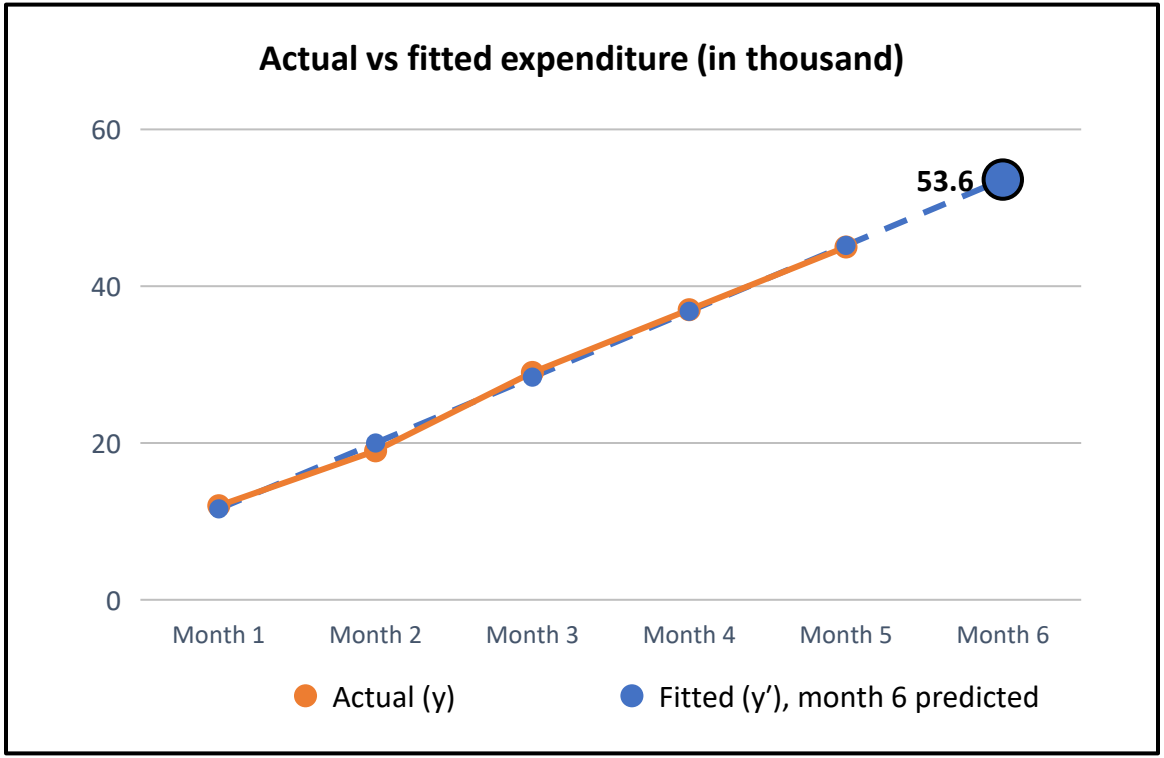
Sr. No	x	y	y′
1	1	12	11.6
2	2	19	20.0
3	3	29	28.4
4	4	37	36.8
5	5	45	45.2
Total	**15**	**142**	**142.0**

Regression line

$$y' = 8.4x + 3.2$$

$\Sigma|e| = 2.4$ and $\Sigma e^2 = 1.6$ are used for performance evaluation.

Prediction for the 6th Month



Estimate for month 6

$y' = 8.4(6) + 3.2$

53.6

thousand expenditure expected

Solution (iii): Fit the Regression Line

Sr. No	x	y	y′	e = y − y′	|e|	e²	|e| ÷ y	Naive error $\|y_t - y_{t-1}\|$
1	1	12	11.6	0.4	0.4	0.16	0.0333	–
2	2	19	20.0	-1.0	1.0	1.00	0.0526	7
3	3	29	28.4	0.6	0.6	0.36	0.0207	10
4	4	37	36.8	0.2	0.2	0.04	0.0054	8
5	5	45	45.2	-0.2	0.2	0.04	0.0044	8
Total	**15**	**142**	**142.0**	**0.0**	**2.4**	**1.60**	**0.1165**	**33**

Regression line: **$y' = 8.4x + 3.2$**

These totals feed the performance evaluation: $\Sigma|e|$, Σe^2, $\Sigma|e| \div y$ and the naive error sum.

Evaluating Our Regression Model

Metric	Calculation (n = 5)	Result
MAE	(0.4 + 1.0 + 0.6 + 0.2 + 0.2) / 5 = 2.4 / 5	0.48
RMSE	√(1.6 / 5) = √0.32	0.566
MAPE	(3.33 + 5.26 + 2.07 + 0.54 + 0.44) / 5 %	2.33 %
MASE	0.48 / 8.25 (naïve MAE = (7+10+8+8) / 4)	0.058

Interpretation: on average the line misses actual expenditure by only 0.48 thousand (about 2.3 %). Since MASE is far below 1, the model beats the naïve baseline, so the month 6 estimate of 53.6 thousand is reliable.